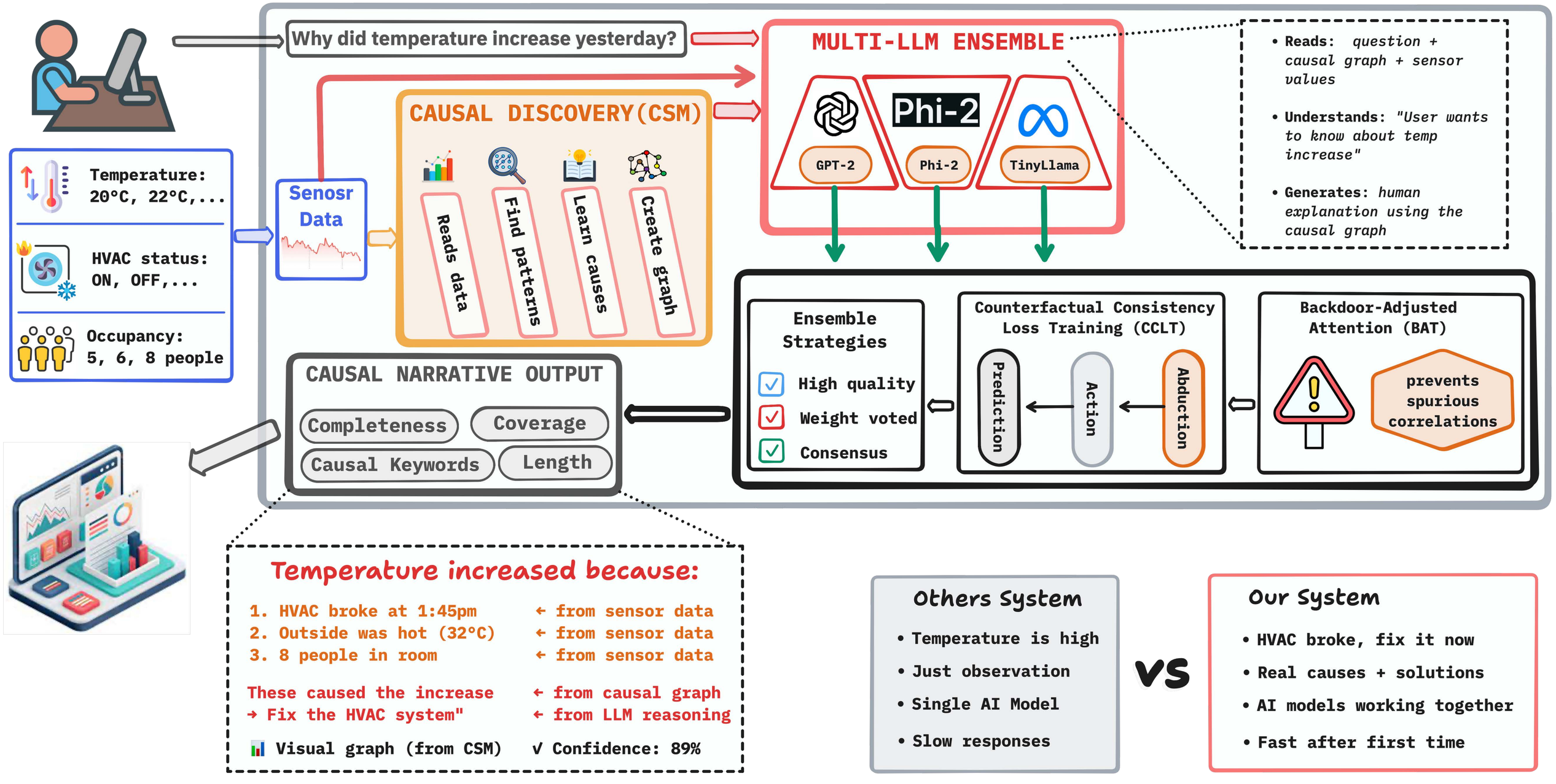
Why did temperature increase yesterday?
Temperature: 20°C, 22°C,...
HVAC status: ON, OFF,...
Occupancy: 5, 6, 8 people
Senosr Data
CAUSAL DISCOVERY(CSM)
Reads data
Find patterns
Learn causes
Create graph
MULTI-LLM ENSEMBLE
Phi-2
GPT-2
Phi-2
TinyLlama
• Reads: question + causal graph + sensor values
• Understands: "User wants to know about temp increase"
• Generates: human explanation using the causal graph
Ensemble Strategies
High quality
Weight voted
Consensus
Counterfactual Consistency Loss Training (CCLT)
Prediction
Action
Abduction
Backdoor-Adjusted Attention (BAT)
prevents spurious correlations
CAUSAL NARRATIVE OUTPUT
Completeness
Coverage
Causal Keywords
Length
Temperature increased because:
1. HVAC broke at 1:45pm ← from sensor data
2. Outside was hot (32°C) ← from sensor data
3. 8 people in room ← from sensor data
These caused the increase ← from causal graph
→ Fix the HVAC system" ← from LLM reasoning
Visual graph (from CSM)
✓ Confidence: 89%
Others System
• Temperature is high
• Just observation
• Single AI Model
• Slow responses
VS
Our System
• HVAC broke, fix it now
• Real causes + solutions
• AI models working together
• Fast after first time